

Angiosperms

Plants that reproduce using flowers are called angiosperms. Their seeds develop with a protective coat. They evolved from gymnosperms about 200 million years ago, and are the most common plant group, at least on land. Unlike the seeds of more primitive plants, angiosperm seeds include a starchy endosperm as a source of nutrition for the growing plant. Wheat, rice, and corn all come from endosperm seeds, and form much of the staple diet of humans.



△ Fruit

Only angiosperms produce fruits. These fruits develop from the outer layers of the ovary after seeds have formed inside. The seeds are often spread by animals, who eat the fruit but cannot digest the seeds.

△ Wood

Tall trees are supported by dead xylem tubes that have been strengthened with a waterproof compound called lignin. The xylem grows out from the center. What remains of the original stem forms the bark.



△ Flower

The flower is the reproductive organ of a flowering plant. Most produce both pollen (male sex cells) and ova, or eggs (female sex cells). The flower is structured to disperse and collect pollen.



△ Leaf

Most angiosperms are broad-leaved. However, plants that live in extreme conditions—such as cacti—have spined leaves to save water loss. Pine needles have the same function.



The **largest flower** belongs to the corpse flower of Southeast Asia—it is 1 m (3 ¼ ft) wide and smells of rotting meat.

Dropping leaves

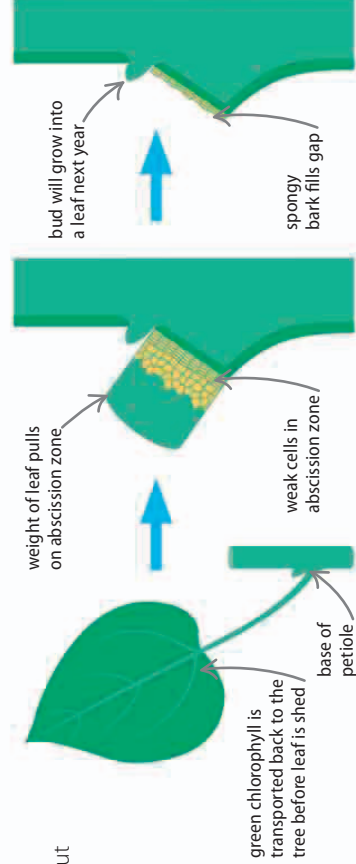
Botanists call the way plants drop their leaves abscission. Deciduous plants drop their leaves all in one go, generally in fall, because there will not be enough sunlight in winter to photosynthesize, and the leaves will be damaged by frost. New leaves grow in spring. An evergreen plant also drops its leaves, but evergreen abscission occurs continually throughout the year, along with new growth.

Many conifer seeds need to be frozen **over winter** before they will sprout.

◁ Evergreen or deciduous?

Evergreen plants live in places that are warm or cold all year, while deciduous species are adapted to habitats with changing seasons.

Climate	Conditions	Which?	Why?
tropical	wet and hot	evergreen	growth possible all year around
monsoon	rainy season	deciduous	avoid water loss through leaves
temperate	cold winter	deciduous	avoid frost damage to leaves
polar	short summer	evergreen	no time to grow new leaves for summer



◁ Abscission

Leaf loss is triggered by changing conditions, such as shortening day lengths. The area at the base of the petiole has thin cell walls. These are broken when spongy bark expands underneath, breaking the water supply to the leaf, so the leaf falls away.